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**Environmental Photochemistry in Hematite-Oxalate System:
Fe(III)-Oxalate Complex Photolysis and ROS generation**

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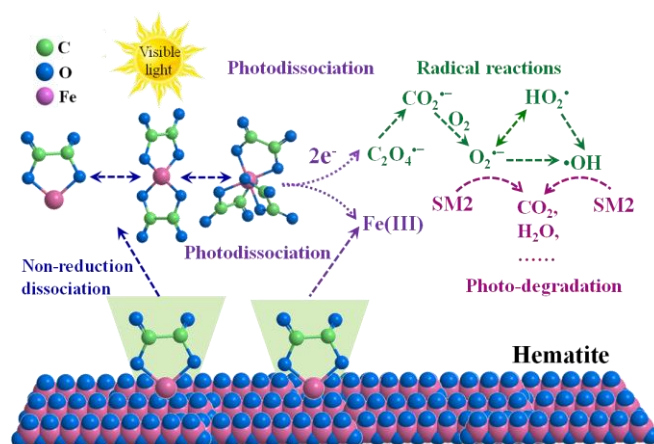
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Graphic Abstract



Highlights

- The interactions between Ox and Ht are investigated by *in-situ* ATR-FTIR
- Fe(III)-Ox complex photolysis is first proved to follow photodissociation mechanism
- No electrons transfer between Fe(III) and Ox during Fe(III)-Ox complex photolysis
- Per Fe(III)-Ox complex generates two $CO_2^{\bullet-}$ radicals in Ht-Ox system
- More $O_2^{\bullet-}$ and $\bullet OH$ are produced in Ht-Ox than that in Fe^{3+} -Ox homogeneous system

Abstract

Oxalates, together with iron (hydr)oxides, play important roles in environmental decontamination by photochemical generation of reactive oxygen species (ROS). However, comprehending reaction mechanism about photolytic behaviors across solid-liquid interface is scarce. Here, the photochemical properties of hematite-oxalate system were investigated from complexation, photolysis, and ROS generation tests. Results implied adsorbed oxalate on hematite surface was detached via non-reductive dissolution by irradiating light. Then, the photolysis of Fe(III)-oxalate complexes on hematite surface or dissolved in solution was followed by photodissociation mechanism, in which per Fe(III)-oxalate complex led to two $\text{CO}_2^{\bullet-}$ formation. Moreover, $\text{O}_2^{\bullet-}/\bullet\text{OOH}$ and $\bullet\text{OH}$ amounts induced by photodissociation mechanism were much higher than that by classic intramolecular ligand-to-metal charge transfer mechanism in Fe^{3+} -Ox homogeneous system. The photo-degradation and mineralization of sulfadimidine (5 mg/L) reached 95% and 59% within 2 h using hematite-Ox system. These findings can help in implementing environmental remediation by utilizing iron (hydr)oxides and oxalate in natural environment.

Key words: Hematite; Oxalate; Photodissociation; Interfacial reaction; Reactive Oxygen Species

1. Introduction

Oxalic acid (OA) and its deprotonated form, oxalate (Ox), have strong coordination ability to Fe(III) with the formation of Fe(III)-oxalate complexes, which could be activated by photo irradiation to produce reactive oxygen species (ROS), including $\text{CO}_2^{\bullet-}$, $\text{O}_2^{\bullet-}$ and $\bullet\text{OH}$ [1-5]. The Fe(III)-Ox complex has been testified to own higher photochemical activity than Fenton reaction (Fe(III/II)/ H_2O_2) towards organic pollutant degradation or poisonous metal removal [6-9]. To date, numerous of researchers utilize oxalate to replace H_2O_2 to design photochemical systems for environmental decontamination, including Fe(III)-Ox homogeneous or heterogeneous systems [10-12]. For instance, the heterogeneous photochemical systems composed of Ox and natural iron (hydr)oxides, such as maghemite [8], hematite [11, 13], ferrihydrite [14], magnetite [15], goethite [16], and Fe-containing photocatalysts [6] have been widely reported. Besides the contaminants degradation performance, the coordination configuration, compositions and properties of Fe(III)-Ox complexes on iron (hydr)oxide surface have attracted continued research interest since it is crucial for the generation of ROS.

Recently, several efforts have been made on the study of the interaction between iron (hydr)oxides and Ox from the molecular level mechanism, particularly with Attenuated Total Reflectance Fourier Transform Infrared spectroscopy (ATR-FTIR) [12, 17-19]. Both experimental and theoretical results agree that Ox can form multiple different complex configurations on iron (hydr)oxides surface, including hydrogen-bonded outer-sphere (Ox is hydrogen-bonded to OH groups attached to the

Fe atoms), and several inner-sphere complexes, such as mononuclear bidentate (MB, Ox is bonded to a single Fe atom via two O atoms), monodentate mononuclear (MM, Ox is bonded to a single Fe atom via one O atom) and binuclear bidentate (BB, Ox is bonded to a single Fe atom via two O atoms) [18, 20]. The amounts and configurations of the Fe(III)-Ox complexes formed on iron (hydr)oxides surface greatly affect the photochemical reactions both on the surface of iron oxides and in the aqueous solution, which has a further impact on the efficiency of contaminants degradation. For example, Huang et al. [12] used ATR-FTIR to clarify the properties of surface Fe-Ox complexes on several iron oxides, containing goethite, hematite, maghemite, and magnetite. Combined with DFT results of Borowski et al. [17], they demonstrated that photocatalytic activity of iron (hydr)oxides-Ox system highly depended on the amounts of MB inner-sphere complexes but slightly with the amounts of MM inner-sphere and outer-sphere complexes. Although the importance of surface Fe(III)-Ox complex was realized, its photolysis mechanism of interfacial reaction and dissolved homogeneous reaction in iron (hydr)oxides-Ox system is still not clear.

In our previous study, the adsorption process and photochemical behavior of OA on the surface of ferrihydrite were investigated by *in situ* ATR-FTIR combined with DFT calculation [14]. Results showed that Fe(III)-Ox complex with MB binding geometry was formed on ferrihydrite, which was inclined to dissolve into solution first, followed by photolytic reaction along with the decomposition of Ox and the reduction of Fe(III). Under natural conditions, ferrihydrite is the preliminary phase

during the process of hydrolysis and precipitation of ferric iron in water, which is a metastable and poorly crystalline oxide [21-26]. Eventually, ferrihydrite inclines to transform to a stable and well crystalline oxide of iron under natural condition, namely hematite [27-30]. Taking into account the structural differences between hematite and ferrihydrite, the photochemical behavior of Ox at the hematite/water interface should be different from that in the ferrihydrite system. Moreover, hematite is a major constituent of soils, rocks and the earth's crust [31], and also usually coexists with OA/Ox in natural environment [11]. Numerous works were reported on the degradation of pollutants using hematite-Ox-light system. However seldom investigations were focused on the complexation, dissolution, and charge transfer process in the hematite-Ox system, which are important in understanding the fundamental reaction mechanism and practical application.

The aim of this study is to bring explanations about the mechanism of Fe(III)-Ox photochemical process in hematite-Ox system under visible light irradiation, including: (1) to explore the interfacial reactions for Ox complexation and dissociation on Ht surface; (2) to examine the effects of Ox concentration, pre-adsorption time, and dissolved oxygen on photolysis of Fe(III)-Ox complexes; (3) to investigate the possible photochemical degradation pathway of Fe(III)-Ox complexes both on the surface of hematite and dissolved in solution, and (4) to study the effect on ROS generation.

2. Experimental sections

2.1. Chemicals

Chemicals used in this work were of analytical grade or higher. Acetonitrile and methanol were obtained from Fisher Scientific. OA ($\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$), nitric acid (HNO_3), ammonium fluoride (NH_4F), sodium hydroxide (NaOH), acetic acid (CH_3COOH), 5,5-Dimethyl-1-pyrroline-N-oxide (DMPO), ferric nitrate ($\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$), ammonium acetate ($\text{CH}_3\text{COONH}_4$), hydroxylamine hydrochloride (HONH_2Cl), thymine, benzoquinone, isopropyl alcohol, sulfadimidine (SM2), and 1,10-phenanthroline monohydrate (o-phen) were acquired from Sinopharm Chemical Reagent Co. Ltd., China.

2.2. Hematite preparation

Hematite (Ht) was obtained by calcinating the ferrihydrite in a muffle furnace at 450 °C for 3 h in air. The details for preparing ferrihydrite have been reported in the previous papers [14, 32]. The structural characteristics of obtained Ht including X-ray diffraction (XRD), scanning electron microscope (SEM) and zeta potential in water at various pH were shown in **Fig. S1-S3** and **Table S1**.

2.3. Experiments and analytical methods

2.3.1. Batch experiments

Photochemical experiments were performed in a 500 mL quartz tube equipped with a quartz water-cooling jacket by adding 0.04 g of Ht (0.2 g/L) in a total volume of 200 mL OA (1 mM, pH 3.1) solution. A certain concentration of $\text{Fe}(\text{NO}_3)_3$ solution was used as the iron source in the homogeneous Fe(III)-Ox system. A 300 W Xenon

lamp was applied as the visible light source with a cutoff filter to ensure the light of wavelengths in the range of 420-780 nm ($500 \pm 5 \text{ mW cm}^{-2}$). The photochemical activity of Ht-Ox system was further evaluated by degradation of 5 mg/L SM2. To investigate the contributions of different ROS and Fe(II) to Ox or SM2 degradation, multiple batch tests were employed in the presence of thymine (10 mM, $\text{CO}_2^{\bullet-}$ scavenger) [33, 34], benzoquinone (10 mM, $\text{O}_2^{\bullet-}$ scavenger) [32], isopropyl alcohol (10 mM, $\bullet\text{OH}$ scavenger) [35], and o-phen (100 mg/L, Fe(II) masking agent) [14]. All experiments were performed in duplicate, each being repeated at least twice.

At predetermined times, 3 mL of aliquots were sampled and filtered immediately with 0.22 mm membrane filters. 1 mL of filtrate was analyzed for the concentration of Ox, and the residual filtrate was analyzed the concentration of Fe(II) and total dissolved iron (TFe).

2.3.2. Analysis methods

Ox was measured using high performance anion exchange chromatography (Thermo ICS-900, USA). Fe(II) and TFe was analyzed with a modified ferrozine method, and SM2 analysis was performed on HPLC (Ultimate 3000, Thermo, USA), the details please check our previous paper [14].

X-ray photoelectron spectroscopy (XPS) data were obtained by a K-Alpha spectrometer (Thermo Fisher Scientific, USA). The C1s peak from the adventitious carbonbased was assumed to be 284.8 eV as the internal reference for calibration. Open circuit potentials (OCP) in Ht-Ox heterogeneous system and Fe(III)-Ox

homogeneous system were collected on a electrochemical station (CHI660E). Ag/AgCl saturated electrode and glassy carbon electrode were used as reference electrode and working electrode, respectively. All potentials were presented with reference to the standard hydrogen electrode (SHE) scale. The formation of $\text{CO}_2^{\bullet-}$, $\bullet\text{OH}$ and $\text{O}_2^{\bullet-}$ were investigated by Electron paramagnetic resonance (EPR). EPR spectra were collected in the presence of DMPO trapping agent (20 mM) at room temperature by a JES FA 200 X-band spectrometer (JEOL, Japan) with a center field of 3231 G, resonance frequency of 9.77 GHz, microwave power of 2.00 mW, modulation frequency of 100 kHz, sweep width of 100 G.

3. Results and discussions

3.1. The Ox complexation and dissociation behavior on Ht surface in dark

At pH 3.1, Ox exists in the solution with the form of HC_2O_4^- and $\text{C}_2\text{O}_4^{2-}$ [14], and Ht presents a positive surface (**Fig. S3**) with enormous surface OH groups (1.42 site/nm², **Table S1 and Fig. S4**). Hence, Ox could attach to Ht surface through electrostatic attraction/hydrogen bonding (out-sphere), or ligand exchange by replacing surface OH groups and forming direct bonds to Fe on Ht surface (inner-sphere, **Fig. S5**). In order to make it more clearly, herein, the interfacial reaction of Ox adsorbed on Ht/water interface and subsequent detachment process were investigated by the time dependent evolution *in situ* ATR-FTIR spectra (**Fig. 1**). The *in situ* ATR-FTIR spectra were gotten by a method similar to that in the previous reports [14, 36], as described in Supplementary Material.

As presented in **Fig. S6**, the homogenous Ox (1 mM) did not affect the FTIR signal of adsorbed Ox on Ht surface. In the Ht-Ox system, **Fig. 1a** presents increasing absorbance in the first 80 min in the three ranges (1) 1250–1320 cm^{-1} , (2) 1380–1470 cm^{-1} , and (3) 1600–1720 cm^{-1} , which imply the increased amount of Ox adsorbed on Ht surface. The peaks at 1709 and 1687 cm^{-1} are attributed to the $\nu_s(\text{C}=\text{O})$ and $\nu_a(\text{C}=\text{O})$, respectively [20]; The peak at 1413 cm^{-1} arises from $\nu_s(-\text{C}-\text{O})$ and $\nu(\text{C}-\text{C})$ combination mode, and that at 1272 cm^{-1} is assigned as $\nu_a(-\text{C}-\text{O})$; those at 1309 and 1624 cm^{-1} correspond to $\nu_s(\text{C}-\text{O})$ and $\nu_a(\text{C}-\text{O})$, respectively [17, 18]. Based on previous ATR-FTIR/DFT studies [14, 17, 20], it is concluded that the peak-positions of adsorbed oxalate located at 1624 and 1309 cm^{-1} belong to the formation of hydrogen-bonded outer-sphere. The peaks at 1709, 1687, 1413, and 1272 cm^{-1} are possibly attributed to MB inner-sphere complexes of adsorbed Ox on Ht surface, as compared with MM and BB inner-sphere complexes [14, 17, 20]. Additionally, the spectrum of adsorbed Ox on Ht surface is resemble to that of aqueous $\text{Fe}(\text{C}_2\text{O}_4)^+$ reported by Persson et al. [37], which was located at 1710, 1690, 1400, and 1255 cm^{-1} . This strong resemblance further suggests that inner-sphere surface complex with a mononuclear five-membered ring chelate structure was generated, namely MB geometry. Meanwhile, MB geometry is proposed for the typical inner-sphere complexes structures of Ox adsorbed on iron (hydr)oxides surface, such as ferrihydrite, goethite and hematite [12, 20, 38, 39]. Furthermore, MB complex owns the highest photochemical activity compared with MM and outer sphere binding geometry, which plays a dominant role in organic contaminants degradation [12, 17],

As presented in **Fig. 1b**, the absorbance of Fe(III)-Ox complexes on Ht surface was decreasing at 1624, 1309 cm^{-1} after 80 min, which is represented to out-sphere complex [20]. However, the absorbance of MB complexes showed insignificant change, which is mainly reflected at 1687 cm^{-1} (**Fig. 1c**), indicating the MB complex formed on the Ht surface was more stable than out-sphere one. Previous study on ferrihyrite-Ox system confirmed that the inner sphere complex was detached from the ferrihyrite surface after 40 min [14]. Hence, *in situ* ATR-FTIR result suggests that MB complex on Ht surface is stronger than that formed on ferrihyrite surface, which implies the crystallinity of iron (hydr)oxide affects the detachment behavior of adsorbed Fe(III)-Ox complexes.

The adsorption and desorption behaviors of Ox in the presence of Ht were further investigated by a batch test in dark condition. As presented in **Fig. 2a**, the concentration of Ox in the solution decreased in the first 10 min, but then slightly increased, indicating the adsorption and desorption processes of Ox occurred in the Ht-Ox system. Besides, a clear detachment behavior of adsorbed Ox on Ht surface happened in the form of inner-sphere complex because an obvious Fe leaching (TFe 0.05 mM) was observed (**Fig. 2b**). However, almost no dissolved Fe(II) was detected throughout the experiment, which can be inferred that Fe(III)-Ox complex was detached from the Ht surface as a non-reductive ligand-promoted dissolution style.

3.2. The photochemical behavior of Ox in Ht-Ox system

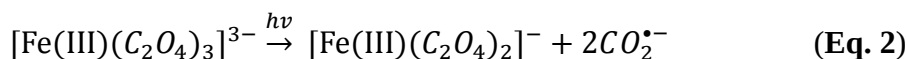
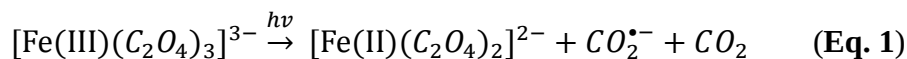
In the absence of Ht, the self-photolysis of Ox was negligible under visible light irradiation (**Fig. 2c**). However, once Ht was introduced, the concentration of Ox was

decreased obviously with the degradation percentage of about 80%, indicating the photochemical decomposition of Fe(III)-Ox complex. According to the pseudo first-order kinetic model of Ox decomposition (**Fig. S7**), the photolysis process of Fe(III)-Ox complex could be divided into two periods. In the first 30 min, Ox was decomposed slowly through an induction period with a 0.0015 min^{-1} pseudo first-order rate constant (**Fig. S7**). Subsequently, the decomposition process of Ox entered into the autocatalytic period, and Ox decomposition rate (0.0164 min^{-1}) is orders of magnitude higher than that of induction period.

However, though Ox was decomposed slowly in the induction period, Fe leaching was significantly fast (**Fig. 2d**) as compared with that in dark condition. The TFe concentration (0.120 mM) in solution was two times higher than that in dark (0.055 mM), implying that the introduction of visible light could act as activation energy to promote the detachment of Fe(III)-Ox complex from the Ht surface. Surprisingly, the dissolved Fe is still presented as Fe(III) in the first 100 min, only slight Fe(II) was detected in the solution at the end of reaction.

For Fe(III)-Ox complex in homogeneous systems under light irradiation, there are two different basic ideas about the mechanism of the primary reactions [33, 40, 41]. One is the intramolecular ligand-to-metal charge transfer (LMCT) mechanism with the production of Fe(II) intermediate as well as CO_2 and $\text{CO}_2^{\bullet-}$ (**Eq. 1**). The other is the photodissociation mechanism, which involves the breakage of an Fe-O bond without electron transfer from the Ox to Fe(III), believed to be $\text{Fe(III)} + 2\text{CO}_2^{\bullet-}$ (**Eq. 2**). Despite a debatable argument on the two pathways, LMCT mechanism is more

accepted since the formation of Fe(II) is observed both in homogeneous iron ions -Ox and heterogeneous iron (hydr)oxides-Ox system [2, 11, 13].



However, in this study, Ox was degraded but without Fe(II) generated in the preliminary stage in the Ht-Ox system. This phenomenon was not in accordance with that reported in other iron (hydro)oxide-Ox photochemical systems. In order to further confirm that the photolysis of Fe(III)-Ox complex in Ht-Ox system follows the photodissociation mechanism, several tests were applied to check the evolution of dissolved Fe(III)/Fe(II) at various conditions.

3.2.1. The effect of Ox concentration

As presented in **Fig. 3a**, Ox could be efficiently degraded in the range of 0.5 to 1.5 mM, and lower Ox concentration led to faster decomposition. After being subjected to visible light for 120 min, about 92%, 75.7% and 73.5% of Ox was decomposed at Ht-Ox systems with 0.5, 1, and 1.5 mM initial Ox concentration, respectively. **Fig. 3b,c** exhibits the dissolved Fe(III) and Fe(II) concentration as a function of time in irradiated Ht suspensions in the presence of Ox at various concentration. Interestingly, no matter what Ox concentration is, nearly zero dissolved Fe(II) was detected in the first 80 min (**Fig. 3c**). The largest amount of dissolved Fe(II) was 0.027 mM while the residual Ox was about 0.09 mM (18%) at 100 min in the 0.5 mM Ox and Ht system. However, when initial Ox concentration increased to 1.5 mM

Ox, nearly zero dissolved Fe(II) was produced in the whole reaction process. This result suggests the dissolved Fe(II) could be merely detected when Ox decomposition enters the terminal stage but not generated in the preliminary stage.

3.2.2. The effect of pre-adsorption

As reported prior studies, the introduction of the pre-dissolution phase could remarkably enhance the initial degradation rate of organic contamination in iron (hydr)oxides–Ox systems, and extending the pre-dissolution time would gradually improve the degradation efficiency of organic contamination [14, 15, 42]. Thus, the pre-dissolution phase would significantly advance the Fe(II) generation.

Not surprisingly, the Ox decomposition rate in induction and automation periods was accelerated remarkably with the pre-adsorption time increasing (**Fig. 3d**). With related to the pre-adsorption time of 0, 60 and 120 min, the rate of Ox decomposition in the period of 0 to 120 min in the system could be calculated as 0.0164, 0.0352 and 0.0435 min⁻¹, respectively (**Fig. S7**). The introduction of the pre-adsorption phase could enhance the detachment of Fe(III)-Ox from Ht surface to dissolve in solution (**Fig. 3e**), which remarkably accelerated the decomposition efficiency of Ox. In addition, zero Fe(II) was detected under visible light irradiation for 50 min in all systems with different pre-adsorption (**Fig. 3f**). However, as expected, the Fe(II) generation was advanced with the introduction of pre-adsorption, and the Fe(II) was produced faster with a pre-adsorption of 120 min than that with 60 min pre-adsorption. This result further proved that Fe(II) was inclined to be formed in the terminal stage of Ox decomposition.

3.2.3. The effect of dissolved oxygen

To obtain a better understanding of the effect of visible light on the dissociation of adsorbed oxalate complexes on Ht surface, the dissolved Fe species were investigated under argon (Ar)-purged condition. **Fig. 4a** exhibits the Fe releasing in the Ht-Ox system under Ar, and no dissolved Fe(II) was generated in dark. Furthermore, the concentration of Fe(III) was almost similar with that in air condition, suggesting Ar has little effect on Fe(III)-Ox complex detaching from Ht surface. However, under visible light irradiation, the Fe leaching presents different tendency between in air and Ar conditions (**Fig. 4b**). In the presence of Ar-purged, the Fe(II) concentration was lower than Fe(III) in the first 20 min, but then Fe(II) further increased with the reaction time increasing. The dissolved Fe(II) concentration after 60 min was even higher than the TFe concentration in air condition, might due to the dissolved Fe(II) could further enhance the iron oxides dissolution via complexation and electron transfer process [14, 43].

Two parallel reaction pathways are postulated for the Fe(II) formation in Ar condition. In pathway 1, the Fe(III)-Ox complex was first detached from the surface of Ht via a non-reductive dissolution style, and then photolysis took place with the generation of $\text{CO}_2^{\bullet-}$. $\text{CO}_2^{\bullet-}$ presents strong reducing activity ($E^0 = -1.97 \text{ V vs. NHE}$) [44], which can react with Fe(III) or Fe(III)-Ox complex to produce Fe(II) [2]. In pathway 2, the photoexcitation of Fe(III)-Ox complex on Ht surface led to the formation of Fe(II) by an LMCT process. Subsequently, the formed Fe(II) on Ht surface was dissolved into solution because of instability of the Fe(II)-O bond [43].

In order to ascertain the pathway of Fe(II) generation, thymine was added into the Ht-Ox system as the $\text{CO}_2^{\bullet-}$ scavenger (**Fig. 4c**). It is surprising that almost zero dissolved Fe(II) was formed, which implies that the generated Fe(II) in solution in the presence of Ar was dominantly produced in the reduction of Fe(III) by $\text{CO}_2^{\bullet-}$, which could further demonstrate that the visible light accelerate Fe(III)-Ox complex breaking away from the Ht surface via non-reductive dissolution.

3.3. The pathway of Ox photolysis in the presence of Ht

3.3.1. Comparative study of Ox photolysis in homogeneous/heterogeneous system

Open-circuit potential measurement is considered as an approximate but simple method for assessing the Fe(III)-Ox complex photolysis pathway from the perspective of potential. Herein, the open circuit potentials in the system of Ht-Ox were recorded, and similar experiment in a homogeneous system composed of 0.15 mM Fe^{3+} and Ox were carried out (**Fig. 5a**). The initial Fe^{3+} concentration used in homogeneous system was slight higher than the maximum TFe leaching concentration in the Ht-Ox system under visible light irradiation (0.126 mM).

In dark, Ht-Ox and Fe^{3+} -Ox homogeneous systems exhibited positive potentials, which was due to both of Fe^{3+} and Ht possess strong oxidization. Under visible light irradiation, the potential shows a trend of decrease in the homogeneous system of Fe^{3+} -Ox, which could be attributed to the drastic rise of Fe^{2+} (reductive) concentration (**Fig. 5b**). Conversely, the potential in Ht-Ox system shows an increase by irradiating visible light. Therefore, it can be speculated that Ox consummation and zero Fe(II) generation in the photolysis of Fe(III)-Ox complex elevated the potential of Ht-Ox

heterogeneous system.

Additionally, the Ox decomposition rate in Ht-Ox system was faster than that in Fe^{3+} -Ox homogeneous system (**Fig. 5c**), though the dissolved TFe in the former was lower than the later, suggesting Ox degradation occurred both in solution and Ht surface. As depicted in **Fig. 5b**, the Fe^{2+} appeared at the moment of visible light irradiation in the Fe^{3+} -Ox homogeneous system, implying that Fe(III)-Ox complex was decomposed via the classic intramolecular LMCT process. However, photodissociation could be attributed to the dominant pathway for Fe(III)-Ox complex decomposition in Ht-Ox system.

3.3.2. Distinguishing homogeneous/heterogeneous decomposition of Ox in Ht-Ox system

In order to investigate the Fe(III)-Ox complexes photolysis mechanism of interfacial reaction, *in situ* ATR-FTIR was applied to characterize the time dependent evolution of surface complexes under visible light irradiation (**Fig. 6a**). For the first step, approximately 30 min was required for Ox adsorption on Ht surface in dark while the spectrum was recorded as the baseline, followed by a photolytic process under visible light irradiation. As presented in **Fig. 6a**, the intensities of absorption peaks of adsorbed Ox decreased step by step during the irradiation by visible light for 120 min. Meanwhile, the intensity of the mode near 1678 cm^{-1} decreased more significantly than that located at 1629 cm^{-1} , which are represented to inner-sphere and out-sphere complexes, respectively. This phenomenon indicates inner-sphere complex was easier to be excited by visible light than out-sphere, which is in accord with other

reports [12, 17]. For example, Borowski et al. [17] found that mononuclear bidentate inner-sphere complex on lepidocrocite was photolyzed at a higher rate than outer-sphere complex under 365 nm UV light irradiation. Huang et al. [12] investigated the role of the Fe(III)-oxalate complexes on iron oxides in determining norfloxacin photodegradation, and the results implied that the degradation efficiency of norfloxacin relies on the amounts of MB inner sphere Fe(III)-Ox surface complex but correlated poorly with the amounts of outer-sphere complexes. In a word, ATR-FTIR result demonstrates Fe(III)-Ox complex on Ht surface could be *in situ* decomposed.

Moreover, the Fe species on Ox-laden Ht before and after illumination were analyzed by XPS (**Fig. 6b**). It clearly shows that the Fe banding energy of neat Ht was located at 711.66 and 710.19 eV, which correspond to Fe(III) [45, 46]. After uptaking Ox, two new peaks appeared at 711.08 and 709.63 eV, respectively, which are ascribed to Fe(III)-Ox complex on Ht surface. Surprisingly, the banding energy of Fe for Ox-adsorbed Ht did not shift after visible light irradiation, which implies that the oxidation state of Fe remained at +3. Besides, the area ratio of Fe(III)-Ox complex decreased from 0.14 to 0.08 and 0.24 to 0.18 for the peaks located at 711.08 and 709.63 eV, respectively, suggesting the amount of Fe(III)-Ox complex decreased after visible light irradiation. This observation is consistent with the results from *in situ* ATR-FTIR. Furthermore, the XPS result reveals that the decomposition of Fe(III)-Ox complex on Ht surface was followed by photodissociation process, in which no electrons transferred from Ox to Fe(III).

To explore the photolysis pathway of dissolved Fe(III)-Ox complex after being detached from Ht surface, the solution was collected after 120 min pre-adsorption with 0.054 mM of dissolved TFe. Then the filter system was subjected to visible light, and a Fe³⁺-Ox homogeneous system consist of 0.06 mM Fe³⁺ and 1 mM OA as the contrast (**Fig. 6c,d**). As surprised, a fast and efficient decomposition of Ox was achieved in filter system, and up to 99% Ox was degraded within 100 min, which almost could rival that in Ht-Ox system (**Fig. 6c**). Furthermore, the Ox decomposition efficiency in the filter system was much higher than that in the Fe³⁺-Ox homogeneous system, in which only 0.2 mM Ox was degraded after 120 min. Particularly, the Fe species evolution during irradiation displays a similar tendency with the Ht-Ox system (**Fig. 6d**), and Fe(II) was not formed in the preliminary stage (within 40 min). Therefore, it can be inferred that in the Ht-Ox photolytic system, photodissociation was the primary process for Fe(III)-Ox decomposition in the preliminary stage. When the Ox was depleted to some extent, intramolecular LMCT was the dominant decomposition process.

3.4. ROS generation

CO₂^{•-} radical is one of the intermediates in the process of Fe(III)-Ox complex decomposition no matter which of LMCT or photodissociation pathway it follows. In the Ht-Ox system with visible light exposure, the generation of CO₂^{•-} radical has been proved by EPR (**Fig. S8**). Subsequently, the generated CO₂^{•-} radicals go on to initiate a reaction of molecular oxygen reduction to produce the O₂^{•-}/•OOH, with a nearly diffusion-controlled rate ($k = 4.2 \times 10^9 \text{ L mol}^{-1} \text{ s}^{-1}$) [47]. In theory, one Fe(III)-Ox

complex molecular decomposition could produce one or two $\text{CO}_2^{\bullet-}$ radicals via intramolecular LMCT and photodissociation pathways (**Eqs. 1, 2**), respectively [40, 41]. Hence, the amount of $\text{O}_2^{\bullet-}/\bullet\text{OOH}$ production in the process of photodissociation should be higher than that in intramolecular LMCT process. Herein, EPR with DMPO as the spin-trapping reagent was employed to investigate the $\text{O}_2^{\bullet-}/\bullet\text{OOH}$ generated in Ht-Ox heterogeneous and Fe^{3+} -Ox homogeneous systems, respectively.

As presented in **Fig. 7a-c**, the four strong characteristic peaks of $\text{DMPO}-\bullet\text{OOH}$ with intensity of 1:1:1:1 appeared in all spectra [14, 48], and another two weak peaks arose from $\text{DMPO}-\text{CO}_2^{\bullet-}$ adduct [2]. Moreover, the intensity of $\text{O}_2^{\bullet-}/\bullet\text{OOH}$ increased with the increasing of irradiating time, implying $\text{O}_2^{\bullet-}/\bullet\text{OOH}$ originates from the photolysis of Fe(III)-Ox complex. As expected, the intensity of $\text{O}_2^{\bullet-}/\bullet\text{OOH}$ in 0.15 mM Fe^{3+} -Ox homogeneous system (**Fig. 7b**) was higher than that in 0.06 mM Fe^{3+} -Ox homogeneous system (**Fig. 7a**), which suggests $\text{O}_2^{\bullet-}/\bullet\text{OOH}$ yield is strongly related to the amount of Fe^{3+} -Ox complex. Nevertheless, the $\text{O}_2^{\bullet-}/\bullet\text{OOH}$ generation in Ht-Ox system (**Fig. 7c**) was superior to that in 0.15 mM Fe^{3+} -Ox homogeneous system, which was almost twice higher than the later by comparing the intensities.

There are two possible reasons for the higher $\text{O}_2^{\bullet-}/\bullet\text{OOH}$ generation in Ht-Ox system. One is the higher amount of Fe(III)-Ox complex photolysis, and the other is the photodissociation of Fe(III)-Ox complex. As discussed above, an initial lag period for the photolysis of Fe(III)-Ox complex exists in the Ht-Ox system, but faster photolysis of Fe(III)-Ox complex in Fe^{3+} -Ox homogeneous system occurs in the preliminary stage (**Fig. 5c**). For example, after 20 min of visible light irradiating,

about 0.03 and 0.13 mM of Ox were depleted in Ht-Ox and Fe³⁺-Ox system, respectively. It means that the yield of O₂^{•-}/•OOH in the Ht-Ox system is higher than that in Fe(III)-Ox homogeneous system, which should be attributed to the pathway of Fe(III)-Ox complex photolysis, not the higher amount of Fe(III)-Ox complex decomposition. The EPR result could well demonstrate that photodissociation of Fe(III)-Ox complex was the dominant reaction at the early period in the Ht-Ox system.

Additionally, spontaneous disproportionation of O₂^{•-}/•OOH then enables rapid production of •OH [6, 8]. Herein, •OH generation was also investigated by EPR (**Fig. 7d-f**). Not surprised, the four-line signal with intensity ratio of 1:2:2:1 known to be DMPO-•OH adduct appeared in all spectra [14, 49, 50], and •OH presented a raising trend with time under visible irradiation. Meanwhile, •OH generation in Ht-Ox system (**Fig. 7f**) was the highest as compared with 0.06 mM (**Fig. 7d**) and 0.15 mM Fe³⁺-Ox homogeneous system (**Fig. 7e**), suggesting that higher O₂^{•-}/•OOH production contribute to higher •OH generation.

3.5. Sulfadimidine degradation

Sulfadimidine (SM2) is one of the most commonly used types of antibiotics in both humans and animals [51]. The wide uses allow its accumulation in the environment, resulting in adverse effects on the ecological systems [52, 53]. Therefore, it is urgent to remediate the wastewater containing SM2 pollutant. Here, SM2 as the contaminant probe was applied to verify the photochemical activity of the Ht-Ox system.

As depicted in **Fig. 8a**, SM2 degradation in neat Ox and Ht-Ox were negligible in dark, suggesting that light activation has a crucial effect on radical production. However, under visible light irradiation, neat Ox or Ht still could not contribute to SM2 degradation. In the presence of Ox and Ht, the efficiency of SM2 degradation was drastically elevated, and 95% of SM2 was degraded, implying Ht-Ox owns remarkable photochemical activity in the removal of organic pollutants. Additionally, the effect on initial SM2 concentration has been investigated, as shown in **Fig. 8b**. 1 mg/L SM2 could be rapidly degraded complete in 20 min, and higher than 99% mineralization of SM2 achieved in 120 min. When the SM2 concentration increased to 5 mg/L, about 59% of SM2 was mineralized with 95% degradation efficiency after 120 min. However, SM2 degradation and mineralization were 76% and 33% within 120 min at an initial concentration of 10 mg/L, still evidencing a high photodegradation activity of the Ht-Ox system.

Meanwhile, the contributions of ROS to SM2 degradation in Ht-Ox system was also carried out via radical capture tests (**Fig. 8c**). When the scavenger of $O_2^{\cdot-}/\bullet OOH$ or $\bullet OH$ was introduced, the SM2 degradation efficiency was drastically dropped, indicating SM2 degradation depends on $O_2^{\cdot-}/\bullet OOH$ and $\bullet OH$. Previous studies show Fe(II) profoundly affects $\bullet OH$ generation [14] and organic degradation [54] in iron (hydr)oxides–Ox systems when Fe(III)-Ox complex photolysis follows LMCT pathway. Herein, the role of Fe(II) on SM2 degradation in Ht-Ox system irradiated by visible light was checked by chelating with o-phen. Interestingly, the presence of Fe(II) masking agent exhibits negligible effect on SM2 degradation, implying that Fe(II)

play a minor role in radical generation in the system of Ht-Ox. This phenomenon is accordant with the result of Fe(III)-Ox complex photolysis in the Ht-Ox system, almost without Fe(II) formed in solution and on Ht surface.

3.6. Proposed reaction mechanisms involved in the Ht-Ox system under visible light irradiation

Based on the results discussed above and prior studies, the Ox photochemical behavior involved in the Ht-Ox system under visible light irradiation was proposed (**Scheme 1**). The Ox was first adsorbed on Ht surface via ligand exchange and electrostatic attraction with the formation of inner-sphere MB and out-sphere complex, respectively. Under visible light, partial MB inner-sphere complex inclined to photolysis *in situ* on the surface of Ht. The residual MB complex tended to detaching from Ht first into solution via non-reductive dissolution with the unique form of $\text{Fe}(\text{C}_2\text{O}_4)^+$. The dissolved $\text{Fe}(\text{C}_2\text{O}_4)^+$ is easily converted to $\text{Fe}(\text{C}_2\text{O}_4)_2^-$ ($k = 6.31 \times 10^6 \text{ M}^{-1}$) and $\text{Fe}(\text{C}_2\text{O}_4)_3^{3-}$ ($k = 3.8 \times 10^4 \text{ M}^{-1}$) step by step [1, 55].

The mechanism of Fe(III)-Ox complex photolysis is open to discussion. The classic view of the Fe(III)-Ox complex photolysis involves intramolecular electron transfer from $\text{C}_2\text{O}_4^{2-}$ to Fe(III) to form Fe(II), followed by the cleavage of oxalate ligands to generate $\text{C}_2\text{O}_4^{\bullet-}$ radical, which dissociates quickly and converts to CO_2 and $\text{CO}_2^{\bullet-}$ (LMCT mechanism) [13, 42]. The other view on the primary Fe(III)-Ox complex photolysis is sequential cleavage of the Fe(III)-O bond (between one oxalate ligand and Fe(III) ion) and the C-C bond of the ligand, in which the oxidation state of

Fe remains constant and $\text{CO}_2^{\bullet-}$ radical is assumed as primary intermediate (photodissociation mechanism) [33, 40, 41].

In homogeneous system, the photolysis of Fe(III)-Ox by irradiating visible light followed the classic pathway, that is, LMCT mechanism, in which Fe(II) was formed with high efficiency. However, in the Ht-Ox heterogeneous system, the dissolved Fe(III)-Ox complex or MB complex on Ht surface was following the photodissociation pathway in the preliminary stage, without electron transfer between Fe(III) and Ox, but showed high production of $\text{CO}_2^{\bullet-}$. Subsequently, the $\text{CO}_2^{\bullet-}$ radicals generated in Ht-Ox system further contribute to $\text{O}_2^{\bullet-}/\bullet\text{OOH}$ generation by donating electrons to dissolved oxygen with a near-diffusion-controlled rate of $2.4 \times 10^9 \text{ M}^{-1} \text{ s}^{-1}$ [2]. Then, $\bullet\text{OH}$ was generated systematically [6, 8]. As attacked by the produced radicals, the photodegradation of SM2 proceeds rapidly and effectively.

4. Conclusion

In this study, the photochemical behavior of Ht-Ox system including complexation, dissociation and charge transfer process were investigated in detail. *In situ* ATR-FTIR spectra suggested the adsorbed Ox on the surface of Ht existed in the form of hydrogen-bonded out-sphere binding geometry and MB inner-sphere binding geometry. Among, MB complex inclined to dissociate from the surface of Ht into solution via a non-reductive process, which could be significantly accelerated by subjecting to visible light. Meanwhile, the MB complexes on Ht surface or dissolved in solution could be activated under illumination, and $\text{CO}_2^{\bullet-}$ radicals were generated through the photodissociation pathway rather than LMCT pathway, in which no

electrons transfer between Fe(III) and Ox. The high production of $\text{CO}_2^{\bullet-}$ radicals further enhanced the $\text{O}_2^{\bullet-}/\bullet\text{OOH}$ and $\bullet\text{OH}$ generation, and the amount of $\text{O}_2^{\bullet-}/\bullet\text{OOH}$ and $\bullet\text{OH}$ induced by photodissociation mechanism was much higher than that by LMCT pathway (Fe^{3+} -Ox homogeneous system). SM2 (5 mg/L) could be fast and efficiently decomposed, 95% degradation and 59% mineralization achieved within 120 min using Ht-Ox-light system. Moreover, Fe(II) masking agent presents a negligible effect on SM2 degradation. Those findings provide comprehensive insights into photolytic behavior in iron (hydr)oxides and oxalate system for environmental decontamination.

CRedit authorship contribution statement

Tianyuan Xu, Xiao Liu, and Lizhi Zhang designed the research. Tianyuan Xu synthesized the hematite, conducted material characterization, and performed the ATR-FTIR, XPS, open-circuit potential, and EPR analysis. Tianyuan Xu, Tianyi Tong, and Yabei Xia conducted Ox complexation, dissociation, and photochemical degradation experiments. Yumin Fang, Tianyi Tong, and Yabei Xia conducted SM2 degradation experiments. All the co-authors co-wrote the manuscript.

Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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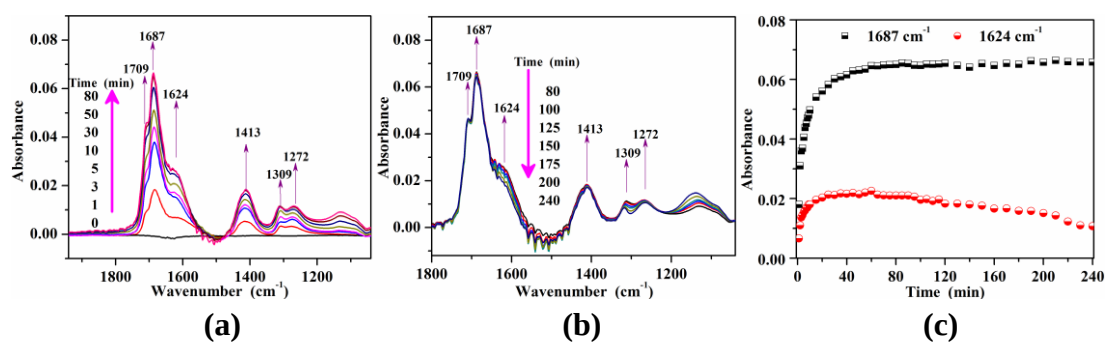


Fig. 1. Time dependent ATR-FTIR spectra of adsorbed Ox on Ht. **(a)** 0-80 min; **(b)** 80-240 min; **(c)** Time dependent evolution of intensity 1624 and 1687 cm^{-1} .

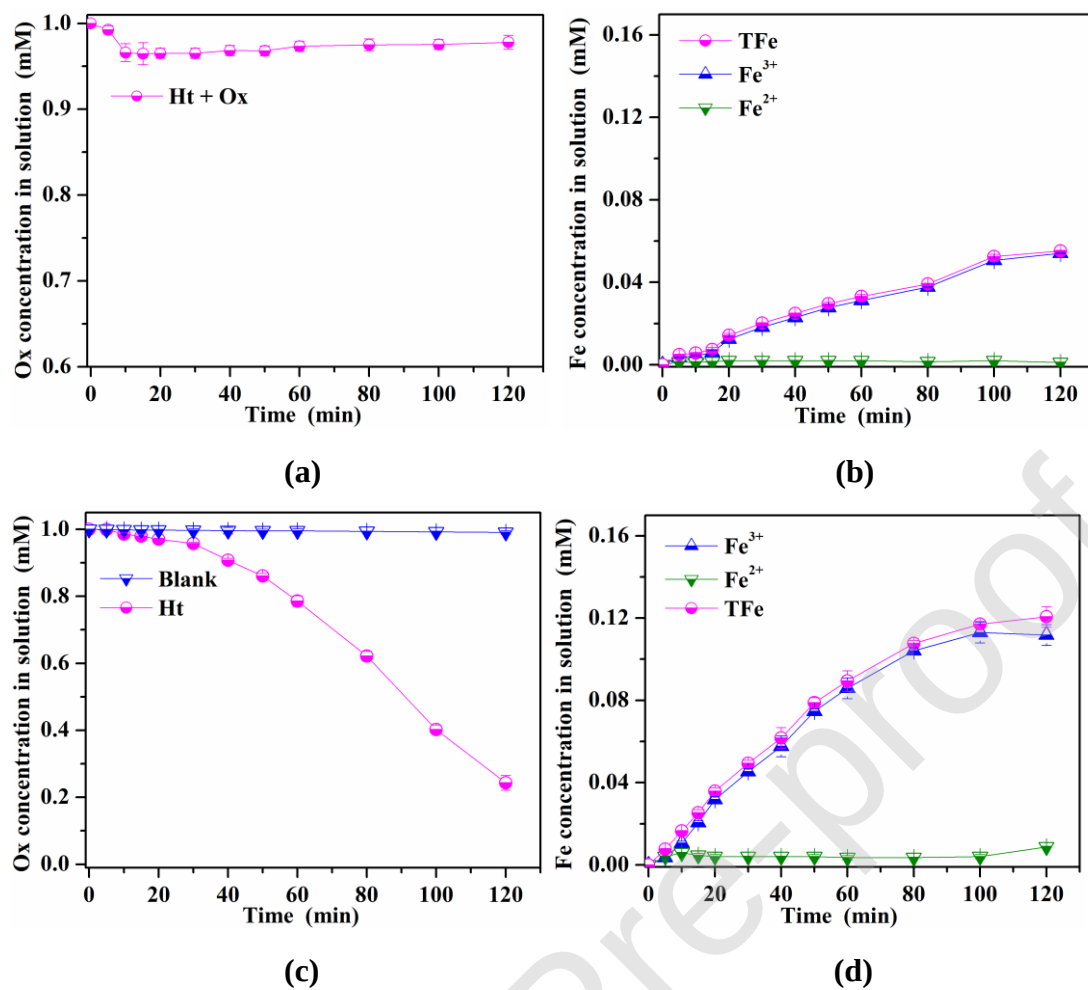


Fig. 2. (a) Time dependent evolution of Ox concentration and (b) Fe released in the solution of Ht-Ox system in dark; (c) Ox degradation kinetics and (d) Fe released in the solution of Ht-Ox system under visible light irradiation. Experimental conditions: Ht = 0.2 g/L, initial Ox = 1.0 mM, and pH = 3.10 ± 0.05.

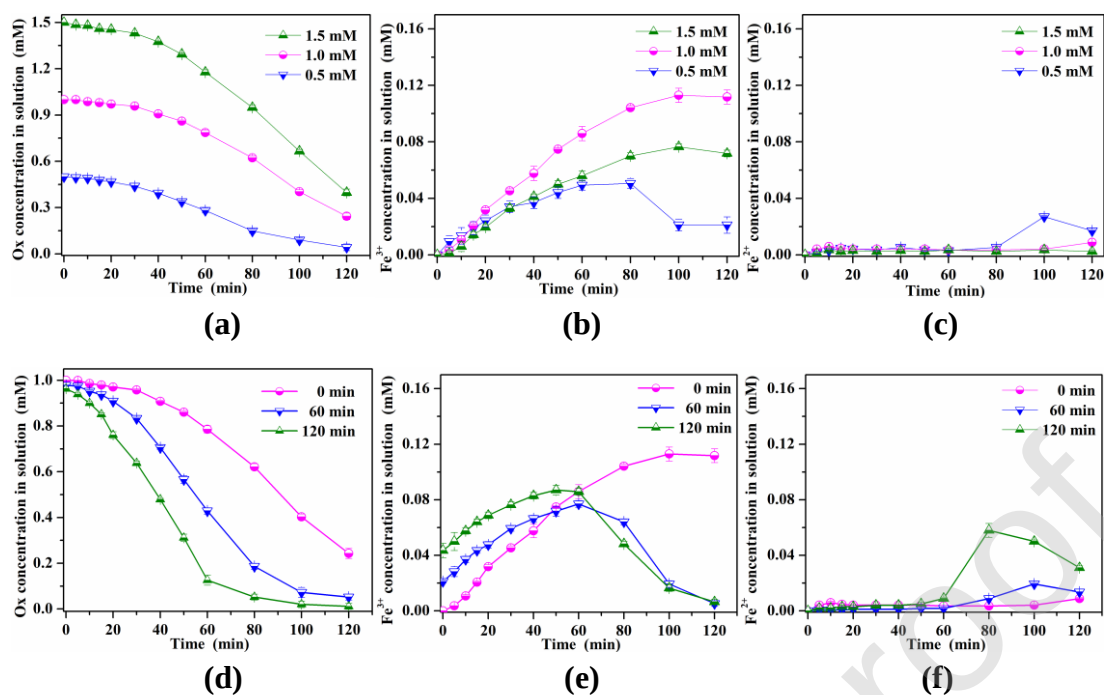


Fig. 3. (a) Ox degradation kinetics, (b) Fe³⁺ and (c) Fe²⁺ released in the solution of Ht-Ox system under visible light irradiation with different initial Ox concentration; (d) Ox degradation kinetics, (e) Fe³⁺ and (f) Fe²⁺ released in the solution of Ht-Ox system under visible light irradiation with different pre-dissolution time. Experimental conditions: Ht = 0.2 g/L, initial Ox = 1.0 mM, pH = 3.10 ± 0.05.

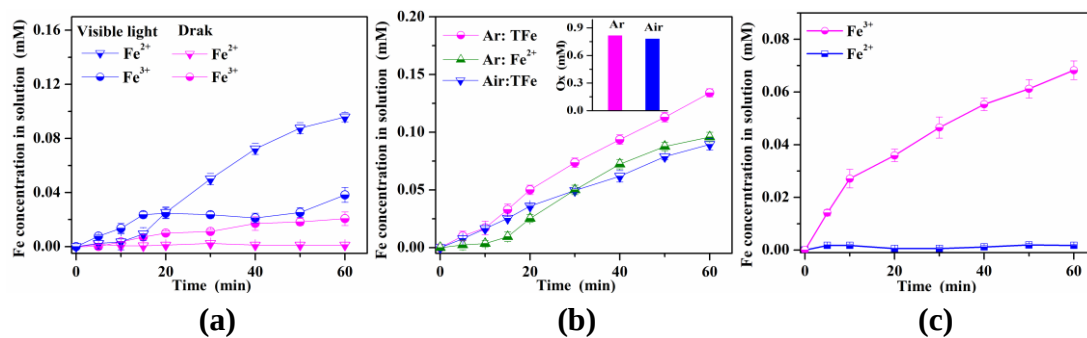


Fig. 4. (a) Fe released in the solution of Ht-Ox system under argon-purged condition; **(b)** Fe released and Ox degradation in Ht-Ox system with argon or air -purged under visible light irradiation; **(c)** Fe released in Ht-Ox system in the presence of thymine and argon under visible light irradiation. Experimental conditions: Ht = 0.2 g/L, initial Ox = 1.0 mM, pH = 3.10 ± 0.05, and thymine = 10 mM.

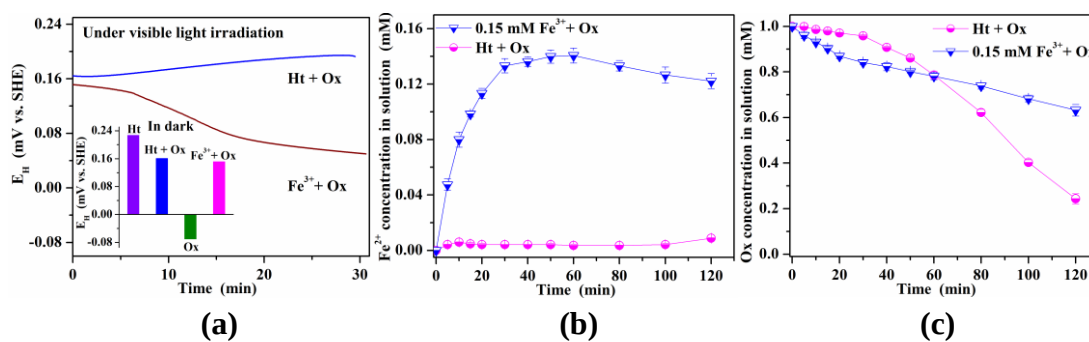


Fig. 5. (a) Open circuit potentials in Ht-Ox heterogeneous system and Fe³⁺-Ox homogeneous system; (b) Fe²⁺ evolution and (c) Ox degradation kinetics in Ht-Ox heterogeneous system and Fe³⁺-Ox homogeneous system under visible light irradiation. Experimental conditions: heterogeneous system: Ht = 0.2 g/L, initial Ox = 1.0 mM, pH = 3.10 ± 0.05; Homogeneous system: initial Fe³⁺ = 0.15 mM, initial Ox = 1.0 mM, pH = 3.10 ± 0.05.

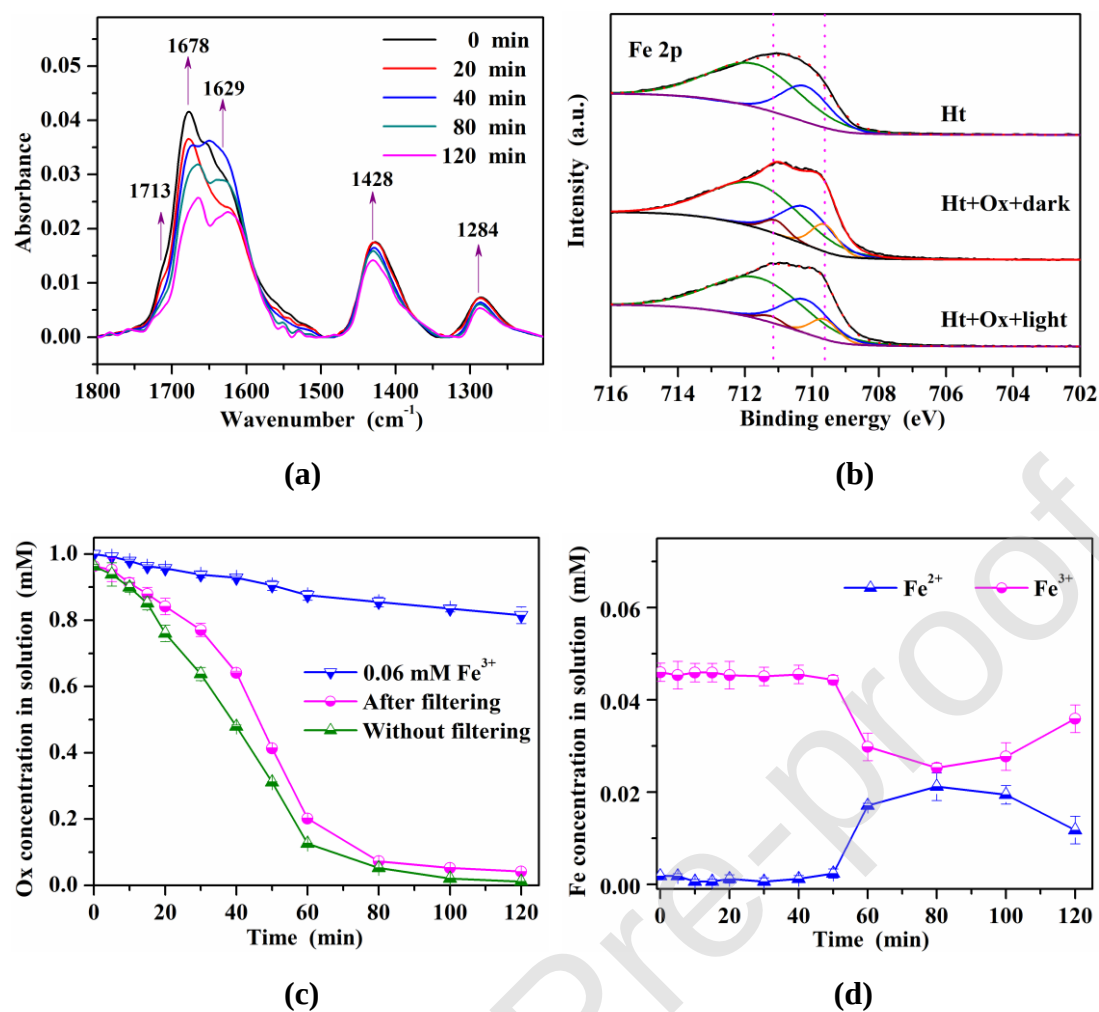


Fig. 6. (a) Time dependent ATR-FTIR spectra of adsorbed Ox on Ht under visible light irradiation. Spectra were collected before irradiation (30 min) and after 120 min of light exposure in static mode (nonflow conditions); **(b)** The Fe 2p_{3/2} XPS spectra of Ht. Time dependent evolution of **(c)** Ox and **(d)** Fe³⁺/Fe²⁺ in the filtering system which was removed Ht after 120 min reaction in dark. The concentration of TFe in the solution of Ht-OA system after 120 min reaction in dark was 0.05 mM.

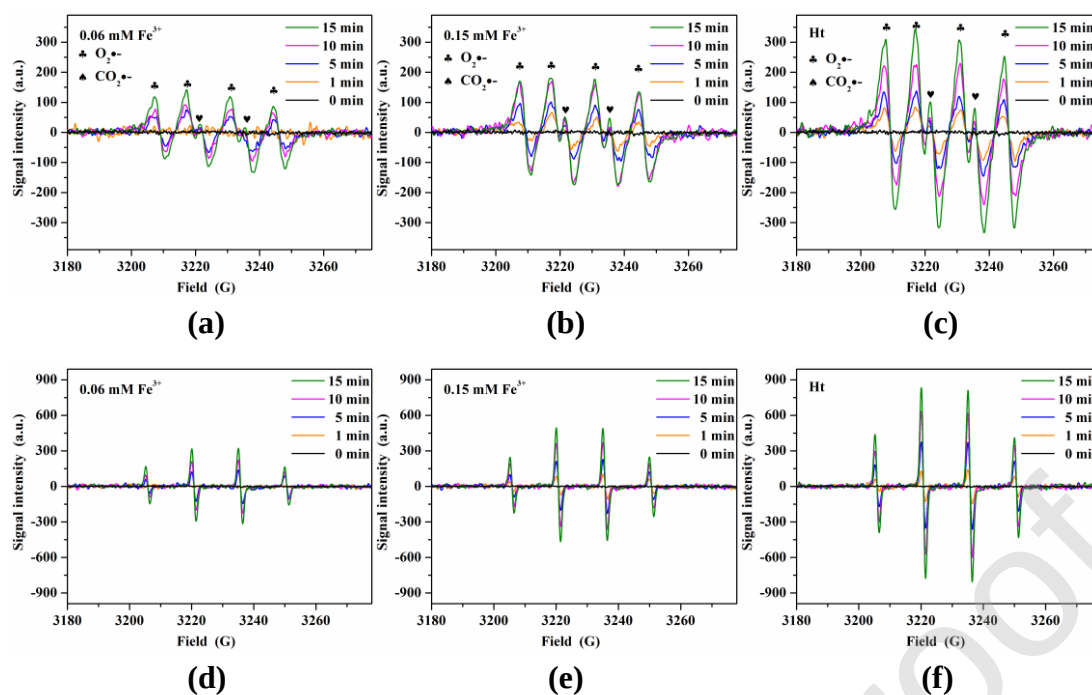


Fig. 7. EPR spectra of DMPO•OOH adducts in **(a)** 0.06 mM, **(b)** 0.15 mM Fe^{3+} -Ox homogeneous system and **(c)** Ht-Ox heterogeneous system under visible light irradiation; EPR spectra of DMPO•OH adducts in **(d)** 0.06 mM, **(e)** 0.15 mM Fe^{3+} -Ox homogeneous system and **(f)** Ht-Ox heterogeneous system under visible light irradiation. Experimental conditions: heterogeneous system: Ht = 0.2 g/L, initial Ox = 1.0 mM, pH = 3.10 ± 0.05 , and DMPO = 40 mM; Homogeneous system: initial Fe^{3+} = 0.06 mM or 0.15 mM, initial Ox = 1.0 mM, pH = 3.10 ± 0.05 , and DMPO = 40 mM.

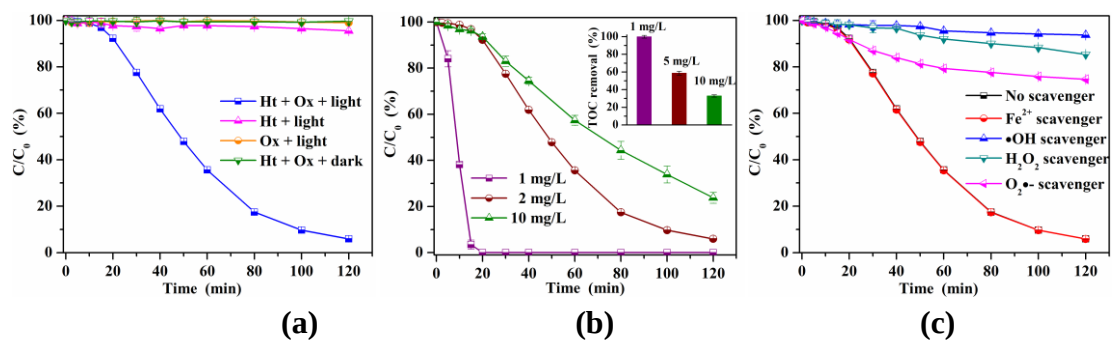
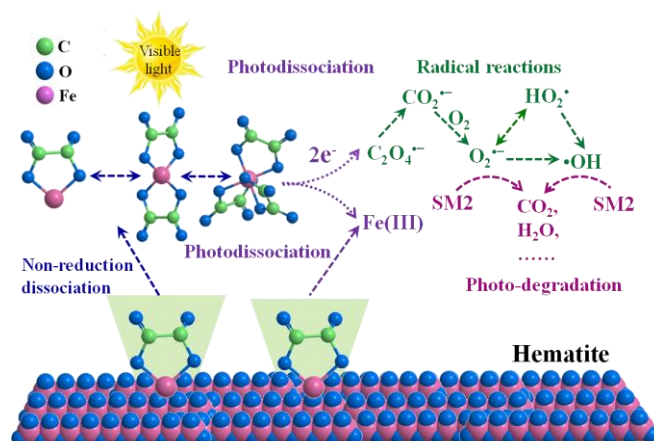


Fig. 8. (a) SM2 degradation in different conditions; **(b)** SM2 degradation in Ht-Ox system with different initial concentration; **(c)** SM2 degradation in Ht-Ox system with different kind of scavengers. Experimental conditions: Ht = 0.2 g/L, initial Ox = 1.0 mM, pH = 3.10 ± 0.05 , and initial SM2 in **(a)** and **(c)** = 5 mg/L.



Scheme 1. Proposed reaction mechanisms involved in the Ht-Ox system under visible light irradiation.